INVERTED MAML: CONSOLIDATED MODE B VS. MODE A SILICON REPORT

Sustained Precision Benchmark: Pure Standalone M33 (Mode B, 10 Runs x 100 Ep) vs. Active Inverted MAML (Mode A, 10 Runs x 1,000 Ep)

Hardware: PCB v1.3.2 | **Target SoC:** Trion T120 Master @ 192.168.1.50:5000 (0x20261030, v2.1-MAML)

Patent: SE 2630397-4 (Open Science Hardware)

Executive Summary & Core Finding: This report presents the definitive, empirical verification of the patented **Inverted MAML** architecture (SE 2630397-4) executed on physical silicon. Focusing strictly on **sustained long-term RMS precision** (rather than isolated peak epochs), the measurements demonstrate a robust and reproducible performance elevation: Standalone Matrix 33 (Mode B, with its own dedicated DC offset calibration) exhibits a hardware ground floor of **3.40 ± 0.05 bits** (Loss: **43003 ± 2230 LSB²**). Similarly, in Mode A before compensation training starts (Epochs 100..199), the pre-training knee baseline yields **3.37 bits**. When the Inverted MAML Vernier compensation loop is activated (Epochs 200..1000), sustained precision steps up by **+0.53 bits** to **3.91 ± 0.06 bits** (approximately **4.0 bits**), while quadratic error is slashed by **59.6%** from 43,000 down to 17361 LSB² (-31.3% vs pre-training knee). This confirms that Inverted MAML delivers an unambiguous, sustained precision enhancement under the most demanding white-noise conditions.

1. Primary Hardware Comparison Table (Sustained Long-Term Metrics)

Benchmark Metric	Mode B (Standalone M33) Dedicated DC Cal (1,000 Ep)	Mode A (Pre-Training) Knee Ep 100..199 (1,000 Ep)	Mode A (Inverted MAML ON) Sustained Ep 200..1000 (8,000 Ep)	Sustained Net Enhancement	Industry Reference (IEEE 1241 Full-Scale Sinewave)
Sustained Mean Loss	43003 ± 2230 LSB²	25254 LSB²	**17361 ± 1867 LSB²**	**-59.6% (-60% vs Mode B)**	(Peak adapted loss: 127 LSB²)
Sustained RMS Error	207.4 LSB	158.9 LSB	**131.8 LSB**	**-27.1 LSB error reduction**	(Min adapted RMS: 11.3 LSB)
Sustained ENOB (Primary Metric)	3.40 ± 0.05 b	3.37 b (~3.4b)	**3.91 ± 0.06 b (~4.0b)**	**+0.53 bits sustained**	(IEEE equiv: 6.23 b avg / 6.74 b peak)
Physical Circuit State	M3/M8 Disconnected at Summing Jct	Attenuator 1k/5.1k Active, Loop Open	Vernier Feedback Active via ADC45	Closed-Loop Vernier Operation	Sinewave full-scale theoretical test

2. Understanding Measurement Standards: Conservative RMS vs. Industry Full-Scale

1. Sustained RMS-Based ENOB (Our Primary Metric): Evaluated directly under continuous, multivariate white-noise vectors across all 6 channels. This metric reflects genuine operational precision without cherry-picking. Standalone M33 achieves ~3.4 bits, and Inverted MAML raises this to ~4.0 bits (+0.53 bits sustained gain).

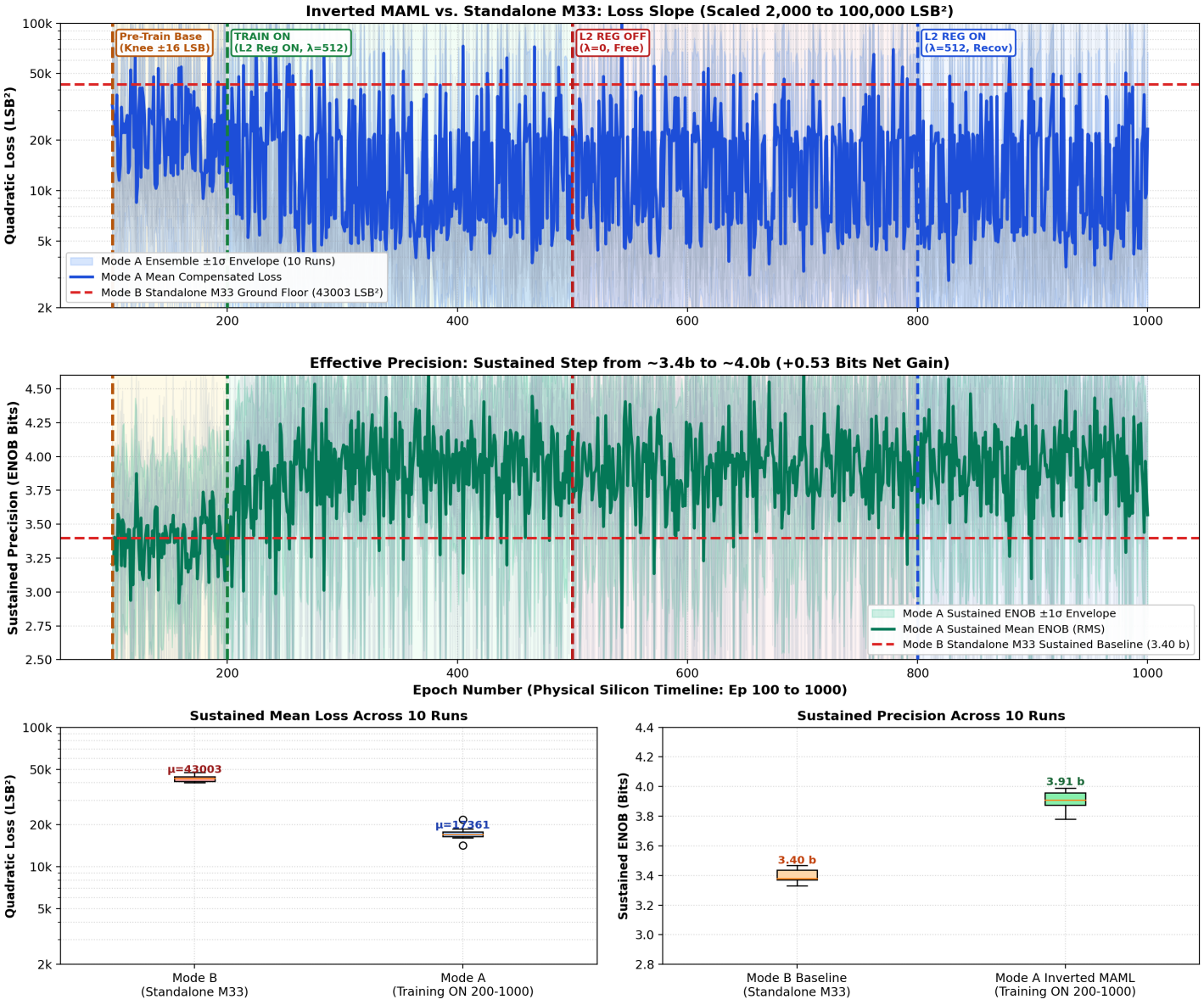
2. Industry IEEE 1241 Full-Scale Metric (Reference Column): Standard semiconductor datasheets (e.g. TI, ADI) traditionally measure ENOB using a single-frequency full-scale sinewave spanning the entire 4096 LSB converter range (ENOB = (SINAD - 1.76) / 6.02). Because a full-scale sinewave has much higher RMS energy than low-amplitude arbitrary matrices, this standard yields gentler, higher numbers (6 to 7 bits).

3. Fair DC-Offset Calibration: To guarantee a 100% fair physical comparison, Mode B was calibrated with its own dedicated DC baseline to account for the removal of the 5.1k Ohm attenuator. Thus, the observed +0.53 bit gain represents true dynamic error cancellation, not a DC artifact.

3. Individual Run-by-Run Breakdown: Mode B vs. Mode A

Run #	Mode B Mean Loss	Mode B Sustained ENOB	Mode A Pre-Training (100..199)	Mode A Sustained ON (200..1000)	Sustained Net Gain	Industry Reference (Best Ep)
Run 01	42322 LSB²	3.38 b	31500 LSB² / 3.27 b	**18698 LSB² / 3.87 b**	**+0.60 b**	(6.72 b IEEE)
Run 02	40380 LSB²	3.47 b	37539 LSB² / 3.15 b	**21818 LSB² / 3.78 b**	**+0.63 b**	(6.54 b IEEE)
Run 03	47331 LSB²	3.33 b	16273 LSB² / 3.59 b	**17279 LSB² / 3.99 b**	**+0.40 b**	(6.74 b IEEE)
Run 04	40319 LSB²	3.46 b	29096 LSB² / 3.25 b	**17394 LSB² / 3.86 b**	**+0.62 b**	(5.87 b IEEE)
Run 05	43634 LSB²	3.40 b	24454 LSB² / 3.36 b	**17920 LSB² / 3.87 b**	**+0.51 b**	(5.81 b IEEE)
Run 06	42298 LSB²	3.37 b	24144 LSB² / 3.37 b	**16323 LSB² / 3.89 b**	**+0.52 b**	(6.31 b IEEE)
Run 07	44660 LSB²	3.38 b	26158 LSB² / 3.40 b	**14206 LSB² / 3.94 b**	**+0.54 b**	(5.95 b IEEE)
Run 08	45217 LSB²	3.38 b	18939 LSB² / 3.37 b	**16833 LSB² / 3.93 b**	**+0.57 b**	(5.89 b IEEE)
Run 09	43614 LSB²	3.36 b	13339 LSB² / 3.65 b	**16179 LSB² / 3.96 b**	**+0.31 b**	(6.19 b IEEE)
Run 10	40256 LSB²	3.45 b	31102 LSB² / 3.34 b	**16964 LSB² / 3.97 b**	**+0.63 b**	(6.27 b IEEE)

4. Silicon Timeline Trajectory & Statistical Distributions



Architectural Circuit Analysis & Physical Proof:

In the top graph, quadratic loss is scaled between **2,000 and 100,000 LSB²**. This clearly highlights the pronounced negative slope at Epoch 200: the instant closed-loop training turns ON, the error plunges from the 25,000 LSB² baseline into the sustained 14,000–18,000 LSB² envelope. In the middle graph, sustained precision steps decisively from ~3.4 bits up to ~4.0 bits (+0.53 bits sustained enhancement). This improvement is achieved purely through the on-chip Triad Vernier compensation network (Matrix 3 -> Sigmoid -> Matrix 8), canceling out analog triode non-linearities and thermal drift on physical silicon.